Electricity

① Charges : $+1.6\times10^{-19}\,C$ / $-1.6\times10^{-19}\,C$ → Coloumb

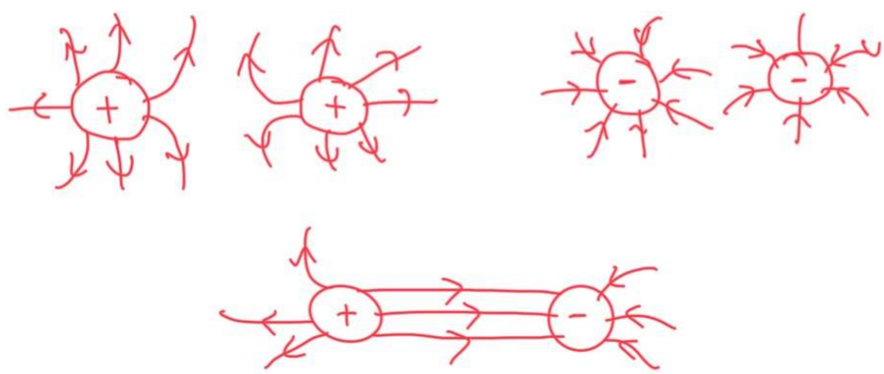
Charge: The number of gained or lost electrons (e^-) times the charge of electrons.

$$Q = n \times 1.6\times10^{-19}$$

Method of charging objects: Rubbing ⇒ By friction

② Electric field: The region around the charge that if any other charge is placed at it, it will be affected. A region where a charge experience a force

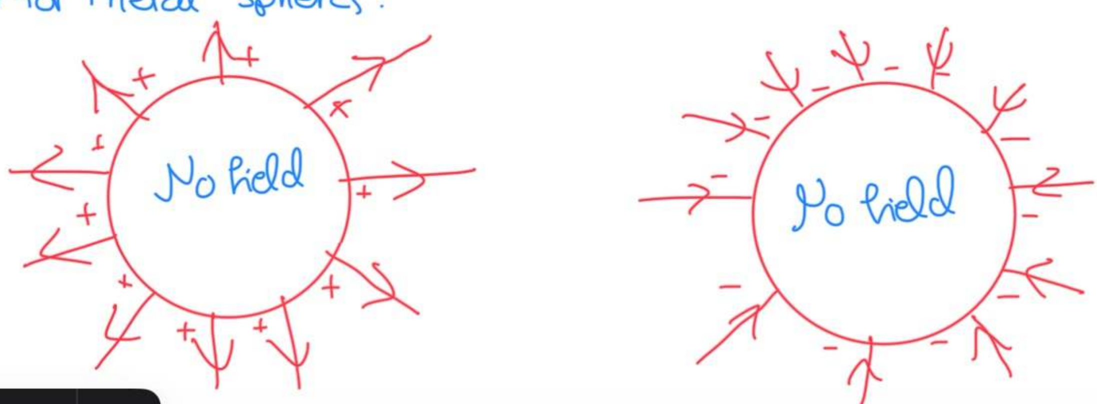
*For point charges:



* For two charged plates:

For stronger field, there will be smaller spaces between chargen

* For metal spheres:



① Types of materials:

- Conductors: Materials that transfer electricity due to the presence of free electrons.

- Insulators: Materials that transfer current poorly due to the lack of free electrons

- Semi-conductors: Insulators that in special situations will be conductors.

② Current: The amount of charge that flows through a point per second.

$$\text{Current} = \frac{\text{charge}}{\text{time}} \qquad I = \frac{Q}{t}$$

$Q \rightarrow C$

$t \rightarrow s$

$I \rightarrow C/s \rightarrow$ Ampere (A)

*Current flows from positive to negative

*Free electrons move from negative to positive.

Potential difference (Pd): The work done by the charges to flow through the circuit per unit charge

$$Pd = \frac{\text{work} \rightarrow J}{\text{charge} \rightarrow C}$$

$Pd \rightarrow J/C \rightarrow$ Volt (V)

Electromotive force (emf): The work done by the source to push the electrons through the circuit per unit charge.

Ohm's law: The current that flows in a wire is directly proportional to voltage across it at constant temperature.

$V \propto I$

$\boxed{V = RI} \rightarrow$ Ohm's law

R: Resistance:
- → The rate between the voltage and the current
- → It's represented by the collisions between the electrons inside the component.
- → It's how hard is the flow of the current
  - * low resistance = high current → Conductors
  - * high resistance = low current → Insulators

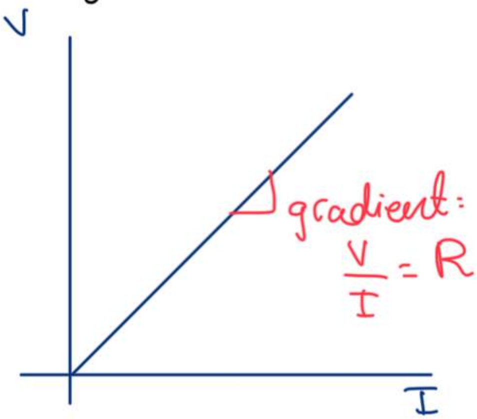
$$R \rightarrow \Omega = \frac{V \rightarrow V}{I \rightarrow A}$$

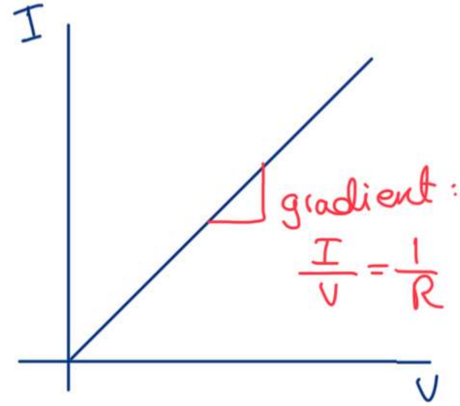
$$I = \frac{V}{R}$$

The voltage is directly proportional to the resistant at constant current.

The current is inversely proportional to the resistant at constant voltage.

Ohmic materials: Example: Metals, materials that obey Ohm's law.

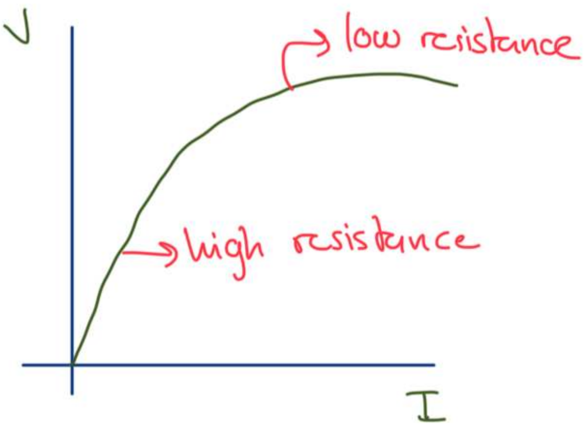


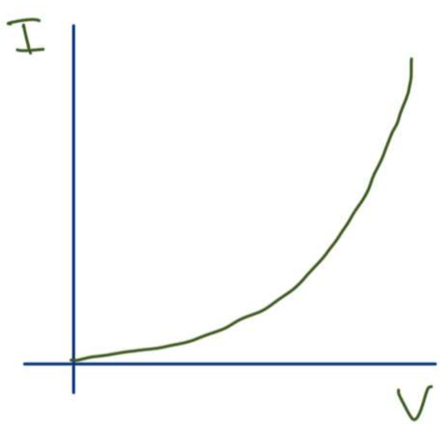


Non-ohmic materials: materials that do not obey ohm's law about resistance

① Thermistor: Depends on temperature 

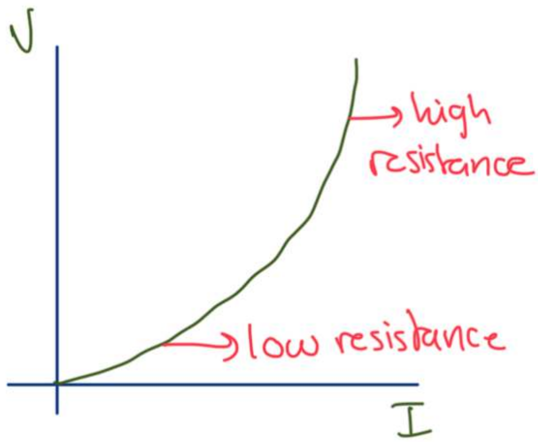
* When hot: Low resistance
* When cold: High resistance

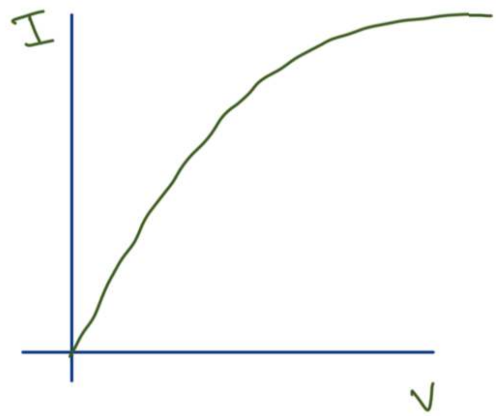




The higher the current → More collision between electrons → more heat → less resistance → less gradient.

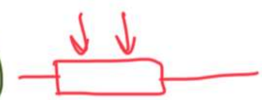
(2) Bulb filament:





The heat increases, so the kinetic energy for atoms and free electrons increases, so more collisions, so more resistance

(3) Light dependant resistance: (LDR)

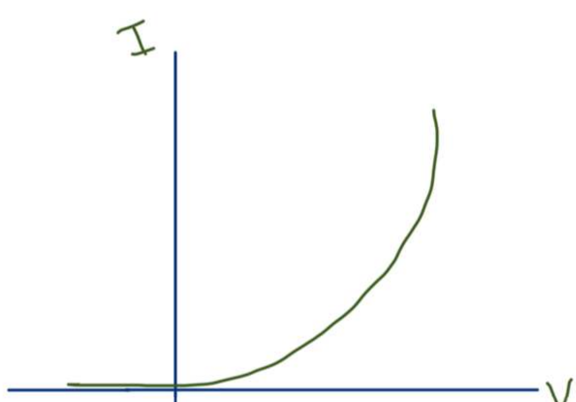


When placed in dark: high resistance

When placed in bright: low resistance

(4) Diode:

Allows the current to pass in one way



Resistanc in wire depends on :

1- Length (L) : Resistance is directly proportional to the length of the wire.

→ More length → more atoms to collide with → more resistance.

2- Cross-sectional area (A) : Resistance is inversely proportional to the length of the wire.

→ Thicker wire → Easier for current to flow → less resistance

3- Temperature : Direct relationship.

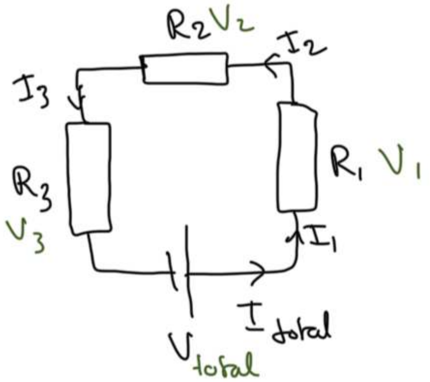
4- Type of material : → Conductor
↳ Insulator

$$\frac{R_1 A_1}{L_1} = \frac{R_2 A_2}{L_2} \quad / \quad \frac{R_1 (d_1)^2}{L_1} = \frac{R_2 (d_2)^2}{L_2}$$

① Series connections:

* All get the same current:

$I_{total} = I_1 = I_2 = I_3$



* Voltage is split:

$V_{Total} = V_1 + V_2 + V_3$

* Combined resistance: $R_C = R_1 + R_2 + R_3$

Advantage of series circuit:

→ Used to protect components from high voltage.

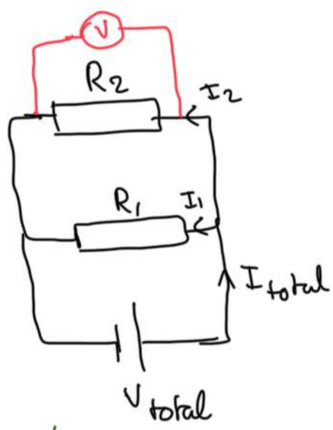
Disadvantage:

→ Any gap in the circuit will terminate the current, so all the circuit is off.

② Parallel connections:

* They all receive same voltage.

$V_{Total} = V_1 = V_2$



* Current is split.

$I_{total} = I_1 + I_2$

* Resistance combined: $\frac{1}{R_C} = \frac{1}{R_1} + \frac{1}{R_2}$

Advantages of parallel series:

1- The resistances all receive equal voltage

2- If any component went off, the others won't be affected

Disadvantage:

→ The battery will end up faster.

Electric power: Electric work done to push the charges per unit time.

$$\text{Power} = \frac{\text{work}}{\text{time}} \qquad P = \frac{W}{t}$$

$W \rightarrow J$

$t \rightarrow s$

$P \rightarrow J/s \rightarrow$ watts : W

$$\text{Voltage} = \frac{\text{work}}{\text{charge}} \qquad V = \frac{W}{Q} \rightarrow W = V \times Q$$

$$P = \frac{V \times Q}{t} \qquad \frac{Q}{t} = I$$

$$P = V \times I$$

$$V = I \times R \rightarrow P = I \times R \times I = P = I^2 \times R$$

$$P = V \times I$$

$$I = \frac{V}{R} \rightarrow P = V \times \frac{V}{R} = P = \frac{V^2}{R}$$

To calculate energy:

Energy = Power x time

$E = P \times t$

$\rightarrow E = IV \times t$

$\rightarrow E = I^2 R \times t$

$\rightarrow E = \frac{V^2}{R} \times t$

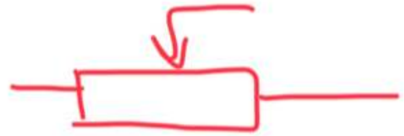
Unit of energy:

$kW.h = 3.6 \times 10^6 J$

$k \rightarrow 1000W$

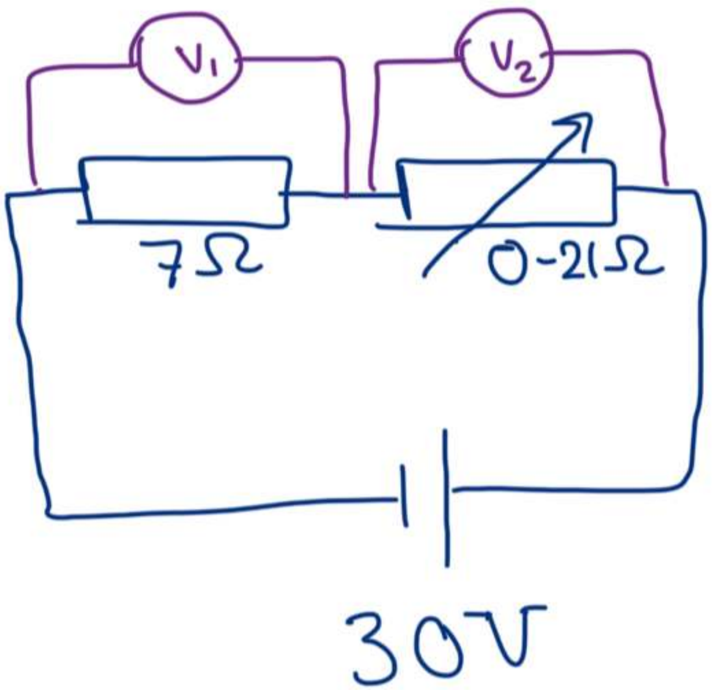
$h \rightarrow 3600s$

$= 3.6 \ MJ$

① Potential divider: 

A series circuit with at least two resistors which are used to divide the voltage.

* Use a variable resistor: 



R_{VR}	V_1	V_2
if: R: 0	30	0
if: R: 7	15	15
if: R= 14	10	20
if: R= 21		

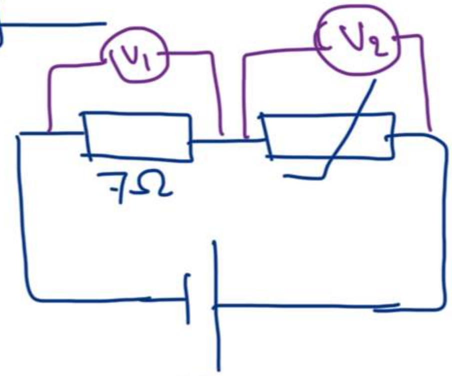
To find V_1 or V_2:

$$V_1 = \frac{V_{Total}}{R_1+R_2} \times R_1 \qquad V_2 = \frac{V_{total}}{R_1+R_2} \times R_2$$

* Use thermistor: 

Hot: low resistance ≈ 0

Cold: High resistance $\approx \infty$



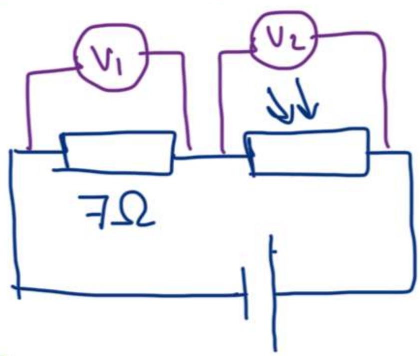
30V

Temperature	R_{Th}	V_1	V_2
Hot	0	30	0
Cold	∞	0	30

* Use light dependent resistor: (LDR): 

Bright: low resistance ≈ 0

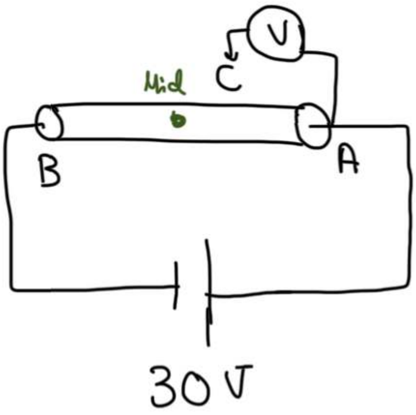
Dim: High resistance $\approx \infty$



30

Brightness	R_{LDR}	V_1	V_2
Dark	∞	0	30
Bright	0	30	0

(2) Potentiometer:

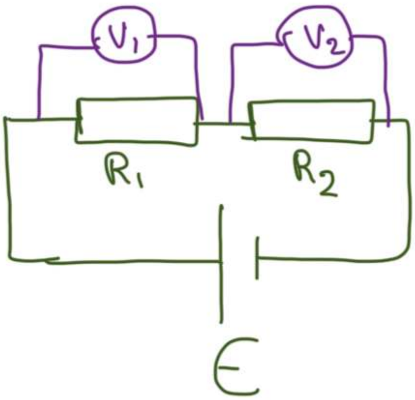


- If C is at A: $V = 0\,V$
- If C is at B: $V = 30\,V$
- If C is at mid point: $V = 15\,V$
- If C is at 20 cm from A:

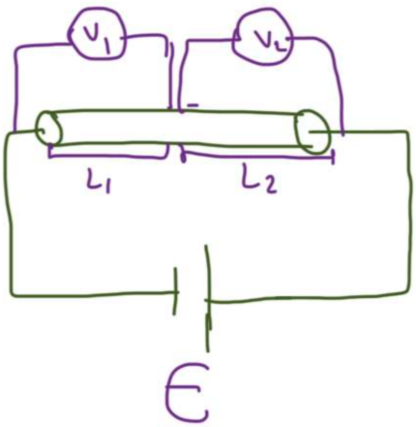
$$\frac{20}{100} \times 30 = 6\,V$$

- If C is at 70 cm from A:

$$\frac{70}{100} \times 30 = 21\,V$$



$$\frac{\epsilon}{R_1 + R_2} = \frac{V_1}{R_1} = \frac{V_2}{R_2}$$



$$\frac{\epsilon}{L_1 + L_2} = \frac{V_1}{L_1} = \frac{V_2}{L_2}$$

Life wire : Carries a current between a negative and positive voltage. (Brown : Input of current)

Earth wire: A wire that connects the metal case of the device with the ground. It makes it safer to touch appliances if it develops a fault. (Green)

Neutral wire: Completes the circuit. It keeps a zero voltage by the electricity company. (Blue : Output of current

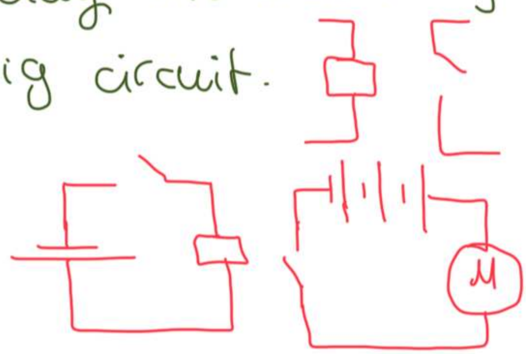
* Fuse: 

A component that is attached at live wire to protect the device from high current. It melts if the current is high, but it's a normal wire if the current is normal

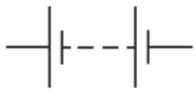
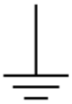
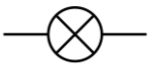
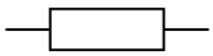
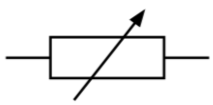
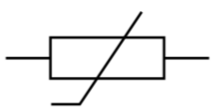
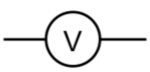
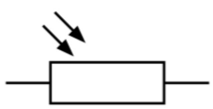
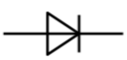
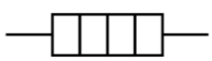
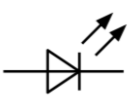
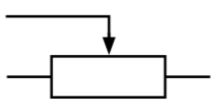
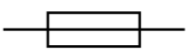
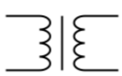
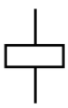
* Circuit breaker:

An electromagnetic switch that is attached to the live wire. If the current is normal, the electromagnet is not strong to grap the switch. If the current is high, it will grab the switch.

* Relay: An electromagnetic switch that controls a big circuit.



Electrical symbols

cell		switch	
battery of cells	 or 	earth or ground	
power supply		junction of conductors	
d.c. power supply		lamp	
a.c. power supply		motor	
fixed resistor		generator	
variable resistor		ammeter	
thermistor		voltmeter	
light-dependent resistor		diode	
heater		light-emitting diode	
potential divider		fuse	
transformer		relay coil	
magnetising coil		electric bell	

1- Fission Reactions: The split of heavy nucleus into two smaller nuclei and release energy.

* The mass of the reactant is larger than the mass of the products. The differences causes the release of energy

2- Fusion Reaction: Joining of two light nuclei to form a heavier nucleus and energy